BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Searching and Sorting

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why should a performance explanation go beyond the word faster?

2. A = 1 and B = 0. What is A AND B?

A) 1 B) 2
C) 0 D) 10

3. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

4. RAM is full. Apps keep moving data between RAM and an SSD. Correct both claims. Explain why. (a) Virtual memory needs no secondary storage. (b) Virtual memory is faster than RAM because SSDs hold more.

Today's objectives

Explain the core ideas of paper 2: searching and sorting using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Linear search

Checks elements sequentially, regardless of order.

Binary search

Requires sorted data and repeatedly halves active search range.

Bubble sort

Adjacent comparisons/swaps move large elements right over passes.

Insertion and merge

Insertion grows a sorted prefix; merge sorts sublists before combining.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Linear search	Checks elements sequentially, regardless of order.
Binary search	Requires sorted data and repeatedly halves active search range.
Bubble sort	Adjacent comparisons/swaps move large elements right over passes.
Insertion and merge	Insertion grows a sorted prefix; merge sorts sublists before combining.

I Do - One worked example

Trace bubble-sort first pass of [6,1,4,2].

1. Compare 6/1 swap -> [1,6,4,2].
2. Compare 6/4 swap -> [1,4,6,2].
3. Compare 6/2 swap -> [1,4,2,6].
4. Conclude 6 has reached the end.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Use binary search to find 17 in [3,7,11,17,23].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the comparison that detects an out-of-order pair of file sizes.

```
leftGB = 6
rightGB = 1
if ... then
print("Swap")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Give the array after one bubble pass on [4,3,2,1]. [3 marks]

Q3. State the prerequisite for binary search. [1 mark]

Q4. Explain insertion sort in two stages. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Binary search reliably finds targets in any unsorted list. ☐ T ☐ F
2. Checks elements sequentially, regardless of order. ☐ T ☐ F
3. Requires sorted data and repeatedly halves active search range. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It should identify the relevant characteristic and explain its effect on the stated task. ☐

Q2 C - 0 ☐

Q3 (0,0), (0,1), (1,1), (1,1). X shows the first gate's output, helping work out Q and find errors. ☐

Q4 (a) Virtual memory uses secondary storage when RAM is too full.
(b) Moving data between RAM and slower secondary storage takes time. More storage space does not mean faster access. ☐

Code First - Model

```
leftGB = 6
rightGB = 1
if leftGB > rightGB then
print("Swap")
endif
```

Output: Swap

We Do - Check the model

1. Found index3. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 [3,2,1,4]. ☐

Q3 The list must be sorted. ☐

Q4 Take next key from unsorted section; shift greater prefix elements right and insert key correctly. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: